

Markups and Public Procurement: Evidence from Czech Construction Tenders

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Excerpt: introduction, data, empirical framework, headline results, and conclusion

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Abstract

This paper analyzes the effect of public procurement on firm markups, using a panel dataset of Czech construction firms from 2006 to 2021. Markups are estimated through a structural framework, while biases are addressed using a selection on observables design and models for unobserved factors. Propensity score-based estimations indicate that firm markups increase by approximately 15% during contract years. A temporal analysis, employing synthetic control and matrix completion methods, reveals that treatment effects decline from around 30% in 2006 to 10% in 2021. These patterns are consistent with institutional improvements in the Czech Republic and offer empirical evidence of increasing efficiency in public spending.

Keywords: Models with Panel Data; Firm Behavior; Empirical Analysis; Procurement; Construction

JEL: C23, D22, H57, L74

¹Predoctoral Fellow, Tobin Center for Economic Policy, Yale University (marek.chadim@yale.edu; <https://marek-chadim.github.io>). This paper is my MSc thesis, submitted to the Department of Economics, Stockholm School of Economics (defended December 2024, awarded the highest grade) and awarded First Place in the 32nd Young Economist of the Year competition of the Czech Economic Society, 2025. I am grateful to my thesis supervisor, Jaakko Meriläinen, for invaluable guidance; to Matěj Bajgar, Mitch Downey, and David Schönholzer for their helpful feedback; and to Jiří Skuhrovec of Datlab s.r.o. for providing administrative data on government procurement. This thesis builds on my earlier work at the Institute of Economic Studies, Charles University Prague (<https://dspace.cuni.cz/handle/20.500.11956/184831>), where I accessed firm financial statement data via MagnusWeb. Replication files: <https://github.com/marek-chadim/thesis>. A substantially revised and extended working-paper version—developed during Ph.D. coursework at Yale and presented in Graduate Industrial Organization II (ECON 601)—is available on request. At Yale, I am grateful to my Tobin Center supervisors, Seth Zimmerman and Barbara Biasi, for their mentorship, and to Joseph Altonji, Steven Berry, Xiaohong Chen, Timothy Christensen, Paul Goldsmith-Pinkham, Philip Haile, Charles Hodgson, Christopher Neilson, Katja Seim, and Edward Vytlačil, whose graduate courses shaped the revision; I especially thank Steven Berry and Katja Seim for detailed comments. This 10-page version reproduces, verbatim, the introduction, data, empirical framework, headline results, and conclusion (body pages 2–3, 7–8, 11–12, 18, 31–32 of the paper); the complete 66-page sample, including the related-work review and all omitted figures, tables, and appendices, is at https://marek-chadim.github.io/chadim_writing_sample.pdf.

1 Introduction

Public procurement constitutes approximately 12% of GDP in OECD countries, representing a substantial share of government expenditure (OECD, 2021). Ensuring its efficiency is vital for reducing resource waste and maximizing taxpayer value. Existing literature has examined key issues in public procurement, including discretion, political favoritism, and limited competition. However, much of this research has focused on comparing tenders, with relatively little attention devoted to differences between public procurement and private market outcomes.

This paper fills this gap by constructing a novel dataset and proposing methodological solutions to the identification challenges inherent in studying the dynamics of firm markups as firms enter and exit public procurement markets. To the best of my knowledge, this is the first study to provide credible econometric evidence on the relationship between markups and participation in public procurement. The findings contribute to ongoing debates on public procurement efficiency, market competition, and the implications of government contracts for firm-level market power.

Using the methodology of De Loecker and Warzynski (2012), I estimate markups under the assumption that cost-minimizing producers optimize variable input expenditures. This approach provides plant-level markup estimates without relying on assumptions about firms' competitive behavior in product markets. The results document the evolution of aggregate markups, showing a decline from 40% above marginal cost in 2006 to 30% in 2021, driven primarily by firms at the upper end of the markup distribution.

Next, I estimate the public procurement premium using non-experimental methods under the assumption of unconfoundedness, conditioning on information available prior to firms' engagement in public procurement. To mitigate biases arising from observational characteristics, I apply methodological advancements from Imbens and Xu (2024). Placebo tests, which compare the average markups of treated and control firms before the treated firms receive contracts, and sensitivity analyses benchmarking the omitted variable bias further reinforce the robustness of the findings.

Finally, I employ panel data models for binary interventions in longitudinal settings to estimate the causal effect of public procurement on the markups of government contractors, drawing on Arkhangelsky and Imbens (2024). These methods extend traditional difference-in-differences and two-way fixed-effects models by incorporating factor models, interactive fixed-effects models, and synthetic control approaches. The analysis emphasizes heterogeneity in causal effects and uses valid inference techniques suited to cases with a small number of treated units.

Drawing causal inferences from observational data, even with advanced methods, requires strong assumptions. The lack of natural experiments, instrumental variables, or regression discontinuity designs limits the availability of exogenous variation in this study. To address these challenges, I use a selection-on-observables approach, framed within the potential outcomes framework, to model counterfactual markups for government contractors in the absence of public procurement. The stable unit treatment value assumption (SUTVA) excludes interference between units and the dynamic effects of past treatments and outcomes. However, in competitive public procurement markets, violations of SUTVA can complicate the interpretation of causal effects. Despite this, the study prioritizes internal validity through a design-based balanced panel approach while enhancing generalizability by leveraging the full dataset. Balanced panel estimates, based on selection on observables, indicate that procurement raises contracting firms' markups by an average of 15%, with the full panel data model showing a decline in this effect from 30% in 2006 to 10% in 2021.

Industrial organization models with heterogeneous producers and firm specific markups provide potential explanations for these findings. More productive firms may absorb the costs associated with entering public procurement markets, which could explain why government contractors typically exhibit higher markups. Additionally, contractors may charge higher markups if they supply higher-quality goods produced using higher-quality inputs. Another plausible explanation is that procurement contracts often provide firms with long-term clients, leading to less elastic realized demand and enabling higher markups.

The literature on public procurement also highlights other mechanisms, such as discretion and political favoritism. My findings suggest that the public procurement markup premium has declined over time, consistent with reforms targeting single-bidding practices Titl (2023) and increased monitoring associated with European Union co-financing (Baránek and Titl, 2024).

The remainder of the paper is structured as follows. Section 1.1 reviews related work. Section 2 introduces the dataset and the construction of the public procurement indicator. Section 3.1 documents the distribution of markups and their descriptive relationship with public procurement. Section 3.2 presents the causal results under the unconfoundedness assumption, while Section 3.3 reports findings from panel data analyses. Methodological details supporting these sections are outlined in Appendices A, B, and C. Finally, Section 4 concludes the paper.

2 Data

I utilize a dataset obtained through licensed² access to the MagnusWeb database³, which provides full company accounts for an unbalanced panel of 1,297 firms active in the Czech construction sector. The dataset contains 7,261 observations covering the period from 2006 to 2021. The sample size accounts for the markup estimation requirement of having at least one lag of data. For robustness of the production function estimates, the top and bottom percentiles of firms are trimmed based on the sales-to-cost-of-goods-sold ratio.

In the Czech Republic, public procurement contracts are awarded under nationwide regulations requiring the publication of contract details by procurers. The dataset includes contracts awarded by central, regional, and municipal governments, as well as government-owned enterprises. A private company⁴ corrected and cleaned the raw data to ensure its accuracy. I link firm financial statements with public procurement records to construct a binary indicator of government tender sales. This classification identifies firms that operate exclusively in the private sector, those that enter public procurement, those that exit it, and those that remain as government contractors.

Year	No. Firms	Share Contractors
2006	227	0.38
2007	290	0.33
2008	348	0.32
2009	412	0.32
2010	497	0.28
2011	506	0.26
2012	457	0.33
2013	235	0.41
2014	243	0.40
2015	245	0.48
2016	338	0.51
2017	660	0.57
2018	708	0.55
2019	764	0.53
2020	769	0.55
2021	562	0.53
Total	7,261	0.44

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3 Results

Section 3.1 documents the main patterns of markups in the Czech construction sector and the relationship between markups and public procurement. I validate these results using two treatment effect identification strategies.

Section 3.2 employs a design-based approach relying on unconfoundedness, or selection based on observable financial statement information and pre-procurement markups. In contrast, Section 3.3 focuses on imputing the counterfactual markup evolution for government contracting firms, addressing selection on latent unobservable factors and incorporating synthetic control methods.

3.1 Markups and Public Procurement in the Czech Construction Sector

I use the empirical framework of De Loecker and Warzynski (2012)⁵ to correlate markups with a firm’s public procurement status and evaluate whether they change upon entering public procurement, controlling for input usage and aggregate trends. While I remain agnostic about a specific theoretical model, I draw on various economic mechanisms to interpret the findings. To estimate markups in the context of public procurement, I incorporate a firm’s public procurement status into input demand equations, allowing it to directly affect the law of motion for productivity. After estimating the output elasticity of the variable inputs, I compute the implied markups using the first-order conditions leading to Equation 1.

The Evolution of Markups in Czech Construction Equation 1 defines the markup as the product of the output elasticity, θ , and the inverse of the variable input’s revenue share, $\frac{PQ}{P_v X_v}$. The revenue share is directly observed in firms’ income statements, while the output elasticities are estimated. These elasticities are both firm- and time-specific, which reflect technological differences across firms and changes over time. The average markup is calculated as:

$$\mu_t = \sum_i m_{it} \mu_{it},$$

where m_{it} represents the weight of each firm. I use the share of sales as the weight and compare it with total costs as the input weight. Figure 1 shows the evolution of average sales-weighted and input-weighted aggregate markups in the Czech construction sector.

⁵See Methods A for markup and production function estimation details.

To the best of my knowledge, this relationship has not previously been documented, making these findings a novel contribution. While the coefficients of the control variables are not the primary focus of this analysis, I revisit the correlation between markups and other economic factors later in the paper. The regression is estimated at the manufacturing level with full year–subindustry interaction dummies. Once δ_1 is estimated, I compute the absolute markup difference by applying the percentage difference to the constant term, which represents the average markup for firms that are inactive in public procurement. This calculation is expressed as: $\hat{\mu}_{PP} = \hat{\delta}_1 \exp(\hat{\delta}_0)$. The estimated markup premium in levels is 0.327 with a delta-method standard error of 0.018. The percentage premium parameter estimate, $\hat{\delta}_1$, is 0.149, with a standard error of 0.003.

These results suggest that government contractors, on average, charge 14.9% higher markups than private-sector firms. They align with microeconomic models in which government contractors charge higher markups due to greater productivity, allowing them to outcompete rivals in tender processes. A comparison of the average markups between government contractors and private-sector firms supports this prediction and highlights its potential policy implications. Models with heterogeneous firms suggest that market share reallocates from less efficient producers to more efficient ones, with government contractors expected to exhibit higher productivity. This higher productivity enables them to cover the fixed costs associated with entering public procurement markets. However, because measured productivity is a residual of a sales-generating production function, it likely also captures unobserved differences between government contractors and private-sector firms. As such, caution is warranted when interpreting the public procurement–productivity relationship.

Public Procurement Entry and Markup Dynamics My dataset allows me to examine how markups evolve for firms entering public procurement markets, both before and after becoming government contractors. I focus on three distinct groups of government contractors identified in the sample: starters, quitters, and continuous contractors⁷. To analyze these groups, I estimate the following regression to compare markups before and after public procurement entry and exit and to measure the markup differential for firms that consistently receive government contracts:⁸

$$\ln \mu_{it} = \gamma_0 + \gamma_1 \text{Entry}_{it} + \gamma_2 \text{Exit}_{it} + \gamma_3 \text{Always}_i + \mathbf{b}'_{it} \sigma + \nu_{it}, \quad (3)$$

⁷ $\text{Entry}_{it} = \mathbb{1}\{\Delta pp_{it} = 1\}$, $\text{Exit}_{it} = \mathbb{1}\{\Delta pp_{it} = -1\}$, $\text{Always}_i = \mathbb{1}\{\frac{1}{16} \sum_{t=1}^{16} pp_{it} = 1\}$

⁸I exclude 152 firms that enter and exit public procurement markets more than once during the sample period to ensure cleaner comparisons.

where $\text{Entry}_{it} = 1$ if a firm becomes a government contractor in year t , and $\text{Exit}_{it} = 1$ if a firm stops contracting with the government. The constant term, γ_0 , represents the average log markup for private-sector firms, including those that enter or exit public procurement markets. The primary coefficients of interest are: - γ_1 , which captures the percentage change in markups for starters (firms entering public procurement) between the pre- and post-entry periods. - γ_2 , which measures the impact of public procurement exit on markups for quitters. - γ_3 , which reflects the markup premium for continuous contractors—firms consistently engaged in public procurement throughout the sample period.

I compute the implied markup-level effect of public procurement entry as:

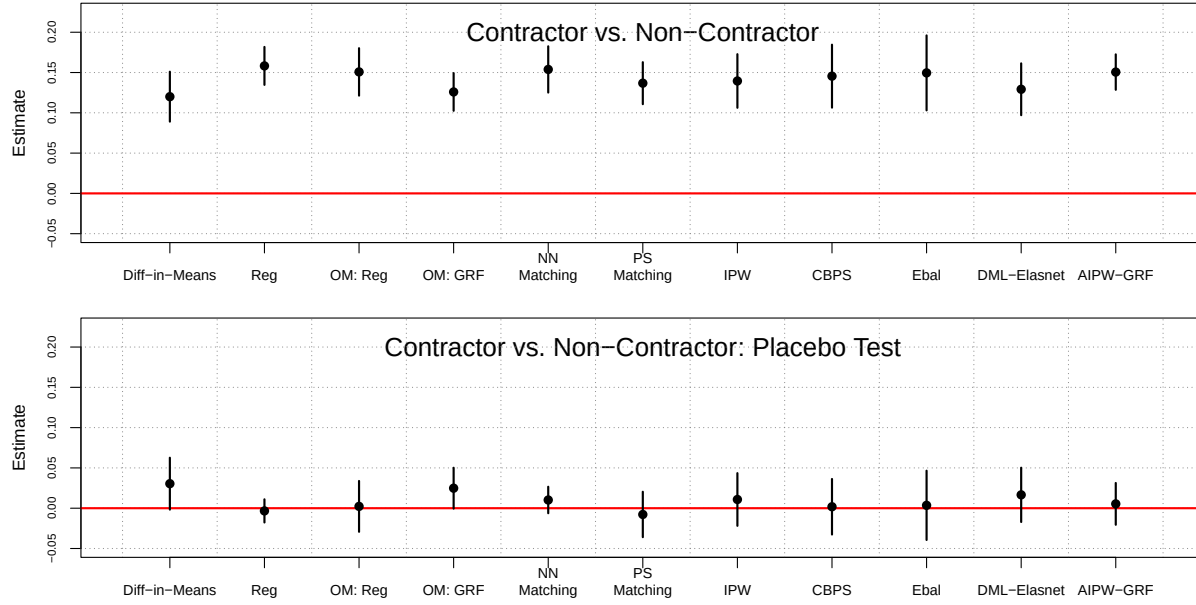
$$\hat{\mu}_{PP}^{\text{entry}} = \hat{\gamma}_1 \exp(\hat{\gamma}_0).$$

My results indicate that public procurement entry is associated with a significant markup increase of approximately 12%, while controlling for aggregate markup dynamics with year and subindustry dummies. All coefficients are reported in Appendix Table 8. The 95% delta-method confidence interval for the level increase ranges from 0.23 to 0.35.

These results suggest that firms entering public procurement experience substantial markup increases, highlighting the role of public contracts in shaping firm behavior. The markup premium for continuous contractors, captured by γ_3 , further suggests that sustained public procurement engagement allows firms to maintain higher pricing power over time.

Discussion This sections reports two main findings: (i) in the cross-section, government contractors exhibit higher markups than their private-sector counterparts within the same industry, and (ii) in the time series, markups increase when firms enter public procurement markets, even after controlling for costs of goods sold, capital, and aggregate demand and supply effects using subindustry-year dummies. Government contracting firms tend to display higher measured total factor productivity, as reflected in the gap between observed revenues and predictions from estimated production functions. More efficient producers may charge lower prices, capture larger market shares in private markets, and outperform rivals in government tenders. Their cost advantage allows them to set higher markups under certain conditions, especially when their relative efficiency translates into competitive advantages in both private and public procurement markets. Alternatively, public procurement markets may differ from private markets in terms of demand elasticities or government valuations of goods and services.

Figure 4: ATT Given Unconfoundedness and Placebo Estimates



Note: The top figure displays ATT (percentage) estimates comparing government contractors to non-contractors for the average annual log markups in Year 0, with 95% confidence intervals. The bottom figure shows ATT estimates for the placebo outcome (average markups from Year -3 to Year -1), with 95% confidence intervals. This analysis employs eleven estimators, including difference-in-means, linear regression, Oaxaca Blinder, GRF as an outcome model, 1:5 nearest neighbor matching with bias correction, propensity score matching, IPW with propensity scores estimated by GRF, CBPS, entropy balancing, double/debiased machine learning with elastic net (using `DoubleML`), and AIPW with GRF (using `grf`).

Table 2: ATT Given Unconfoundedness and Placebo Estimates

Effect on Markups	Contract	Pre-Contract Average
Difference-in-Means	0.12 (0.02)	0.03 (0.02)
Regression	0.16 (0.01)	-0.00 (0.01)
Oaxaca Blinder	0.15 (0.01)	0.00 (0.02)
GRF	0.13 (0.01)	0.03 (0.01)
NN Matching	0.15 (0.01)	0.01 (0.01)
PS Matching	0.13 (0.01)	-0.00 (0.01)
IPW	0.14 (0.02)	0.01 (0.02)
CBPS	0.15 (0.02)	0.00 (0.02)
Entropy Balancing	0.15 (0.03)	-0.00 (0.02)
DML-ElasticNet	0.16 (0.01)	-0.01 (0.01)
AIPW-GRF	0.15 (0.01)	0.00 (0.01)

Note: ATT estimates for the real outcome (markups in Year 0) and the placebo outcome (average markups from Year -3 to Year -1) are based on the trimmed dataset, excluding extreme propensity scores (above 0.8). Robust standard errors are in parentheses. The outcome is measured in natural logarithms and presented as percentage effects.

These estimates are visualized in Figure 4.

Overall, the results lend support to the validity of the identifying assumptions. I specify 2 factors to be extracted from principal components of residuals for the `IFect` model. The `fect` model assumes $\text{rank} = 0$ for factors beyond unit and time fixed effects and slightly outperforms `IFect` in the diagnostic tests. For matrix completion (MC), the optimal λ from cross-validation exceeds 15 of the 16 singular values of the matrix L , whose rank equals the number of periods, which is the minimum of total firms N and periods T . The matrix completion approach employs data-driven regularization to estimate a rank-1 factor model for the outcome, which is residualized from fixed two-way effects in the pre-treatment test sample.¹³ In the test sample, this method reduces residual variation more effectively than both `IFect` and `Fect`, and leads to lower standard errors.¹⁴

I present the year-aggregated ATT estimates from the matrix completion method in Table 4. The results provide further evidence of a decreasing trend in the public procurement premium over time for firms in the Czech construction sector. Specifically, my estimates suggest that, compared to firms operating only in the private sector, government contractors had markups approximately 30% higher from 2006 to 2009, over 20% higher during the period 2010–2015, and about 10% higher during 2016–2021—only one-third of their initial level.

Table 4: Matrix Completion Year-Aggregated ATT Estimates

Year	ATT	Standard Err.	No. Treated
2006	0.294	0.016	12
2007	0.282	0.022	7
2008	0.300	0.014	8
2009	0.337	0.010	18
2010	0.273	0.017	9
2011	0.241	0.012	11
2012	0.236	0.013	18
2013	0.262	0.010	18
2014	0.301	0.008	20
2015	0.219	0.008	31
2016	0.092	0.008	34
2017	0.105	0.008	77
2018	0.100	0.006	68
2019	0.101	0.006	74
2020	0.118	0.007	78
2021	0.098	0.008	58

¹³See Methods C for visualization.

¹⁴Appendix Figure 16 presents the balanced panel equivalent of Figure 15, where specifying the rank with `IFect` outperforms MC in diagnostics, though the treatment effect estimates are larger in magnitude but less precise.

4 Conclusion

Using firm-level data on publicly traded firms in the Czech construction sector, this study employs a method that accommodates diverse price-setting strategies and nonconstant returns to scale to estimate firm-specific markups. It describes the evolution of market power through the distributional characteristics of markups. Between 2006 and 2021, markups declined from 40% to nearly 30%, a decline primarily driven by firms with the highest initial markups. Over time, the distribution of markups became less skewed, with a thinner upper tail.

The decline in Czech construction aggregate markups is linked to the dynamics of the public procurement markup premium. The relationship is validated using policy evaluation designs to account for both observable and unobservable selection bias.

In the balanced sample, the year-of-contract-treatment effect is estimated to be 15% under selection on observable lagged variables, while the same-sample fixed-effects analysis yields a point estimate of 20%. Using both the unconfoundedness approach and the TWFE model to test robustness to alternative identifying assumptions leads to the bracketing property bounds on the estimated causal effect (Angrist and Pischke, 2009, pp. 185).

The analysis also integrates Synthetic Difference-in-Differences re-weighting and matching on pre-exposure trends to reduce reliance on parallel-trend assumptions. This approach accounts for additive unit-level shifts while utilizing data-driven synthetic control methods.

Finally, the full sample estimation employs interactive fixed effects and matrix completion methods. The framework is validated using diagnostic tests of out-of-sample predictions for untreated potential outcomes, which are robust to model misspecification and overfitting. Subgroup analysis indicates that the overall effect is primarily driven by government contracts awarded in the early years of the sample. The estimated average public procurement effect on government contractors is 30% in 2006 and decreases to 10% by 2021.

These findings contribute to the public procurement literature by establishing a relationship between markups and public procurement. They also provide a benchmark for assessing the competitiveness of government projects relative to those in the private sector. The results provide empirical support for reforms aimed at eliminating single-bidding practices and increasing transparency in the tendering process that prevents favoritism in public procurement. They offer policy-relevant insights, highlighting the role of institutional improvements in enhancing the efficiency of public spending.